Decoding New Residential Behavior: GPS-Based Detection of Working from Home and Third Places

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Measuring impact of WFH on cities

Three channels through which remote-work behavior reshapes urban form and public services.

01



Commuting behavior

Less load on public transit

Residents stay home more days per week — reducing peak-hour congestion and freeing capacity on the public-transit network.

02



Residential choice

Re-sorting of where people live

No premium for proximity to the CBD: residents move toward suburbia and amenity-rich periphery, redrawing the rental price gradient.

03



Local development

Growth of neighborhood services

Demand for local cafés, shops, parks and community space rises — triggering new openings, mixed-use zoning and infrastructure upgrades.

Li, W., Zhang, E., & Long, Y. (2024). Unveiling fine-scale urban third places for remote work using mobile phone big data. *Sustainable Cities and Society*, *103*, 105258. https://doi.org/10.1016/j.scs.2024.105258
Gokan, T., Kichko, S., Matheson, J. A., & Thisse, J.-F. (2025). *Teleworking, local labor markets, and the structure of cities*.

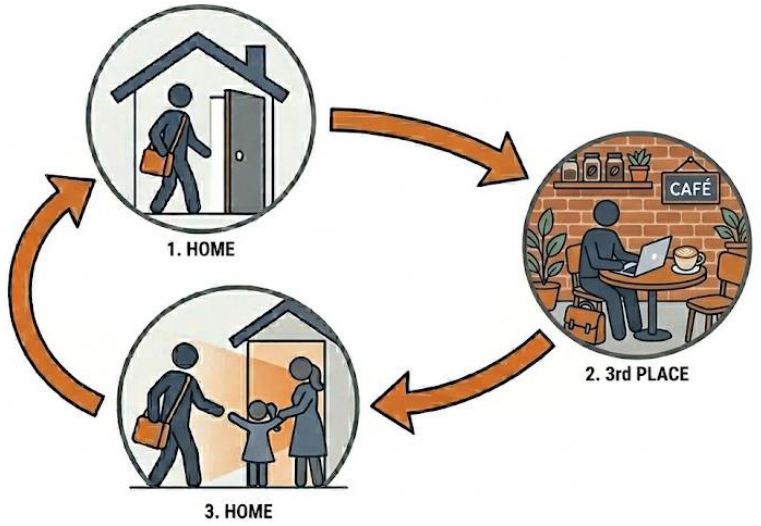
New residential behavior — how to measure?

After COVID-19 the daily cycle no longer runs through the office. People work from home — and part of that time from a third place: a café, a library, a community center.

WORK FROM HOME

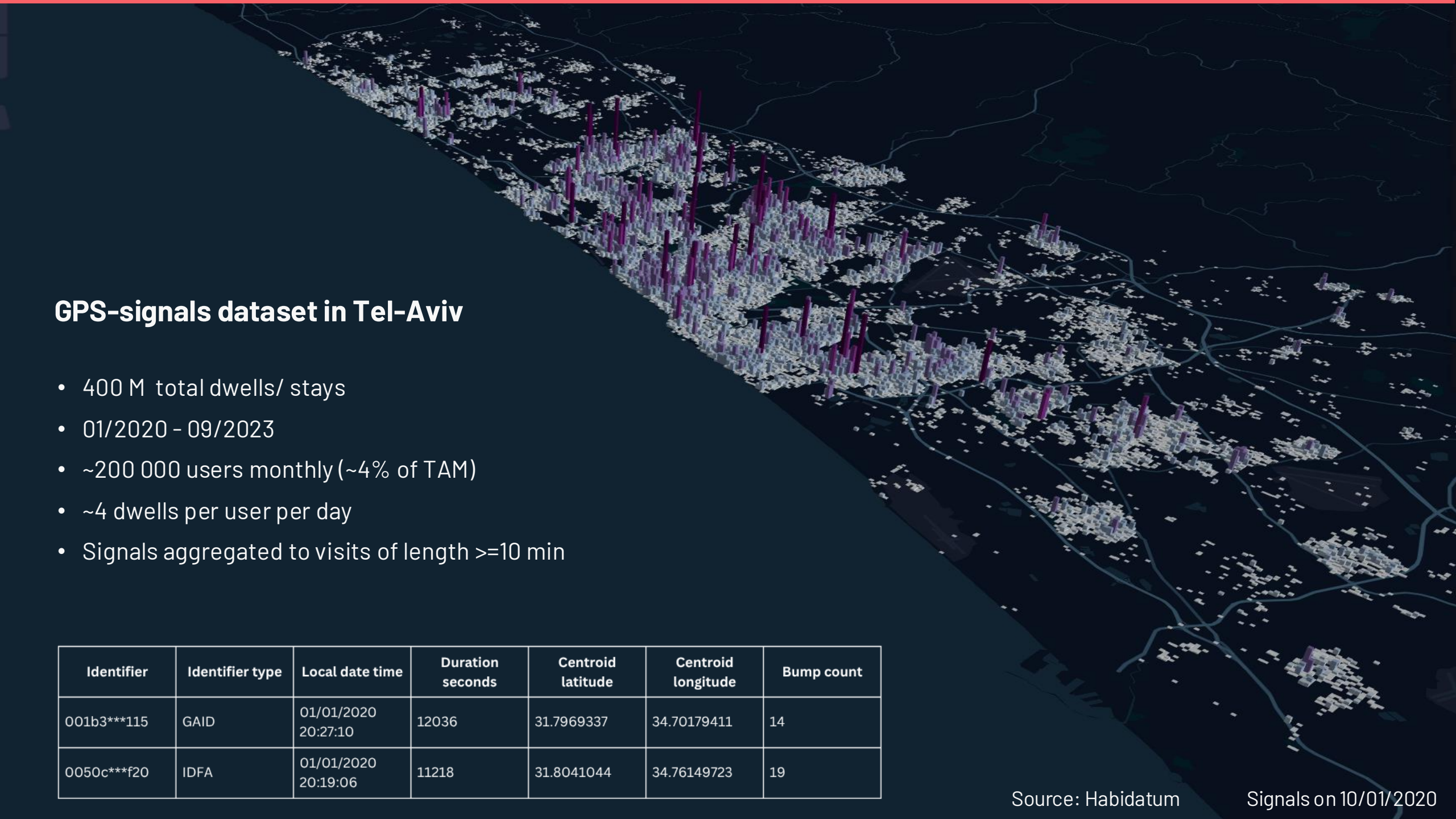


WORK FROM THIRD PLACE



RESEARCH QUESTIONS

- *How common is this new behavior?*
- *Does it persist two years after COVID-19?*
- *How can we detect it in the data?*



GPS-signals dataset in Tel-Aviv

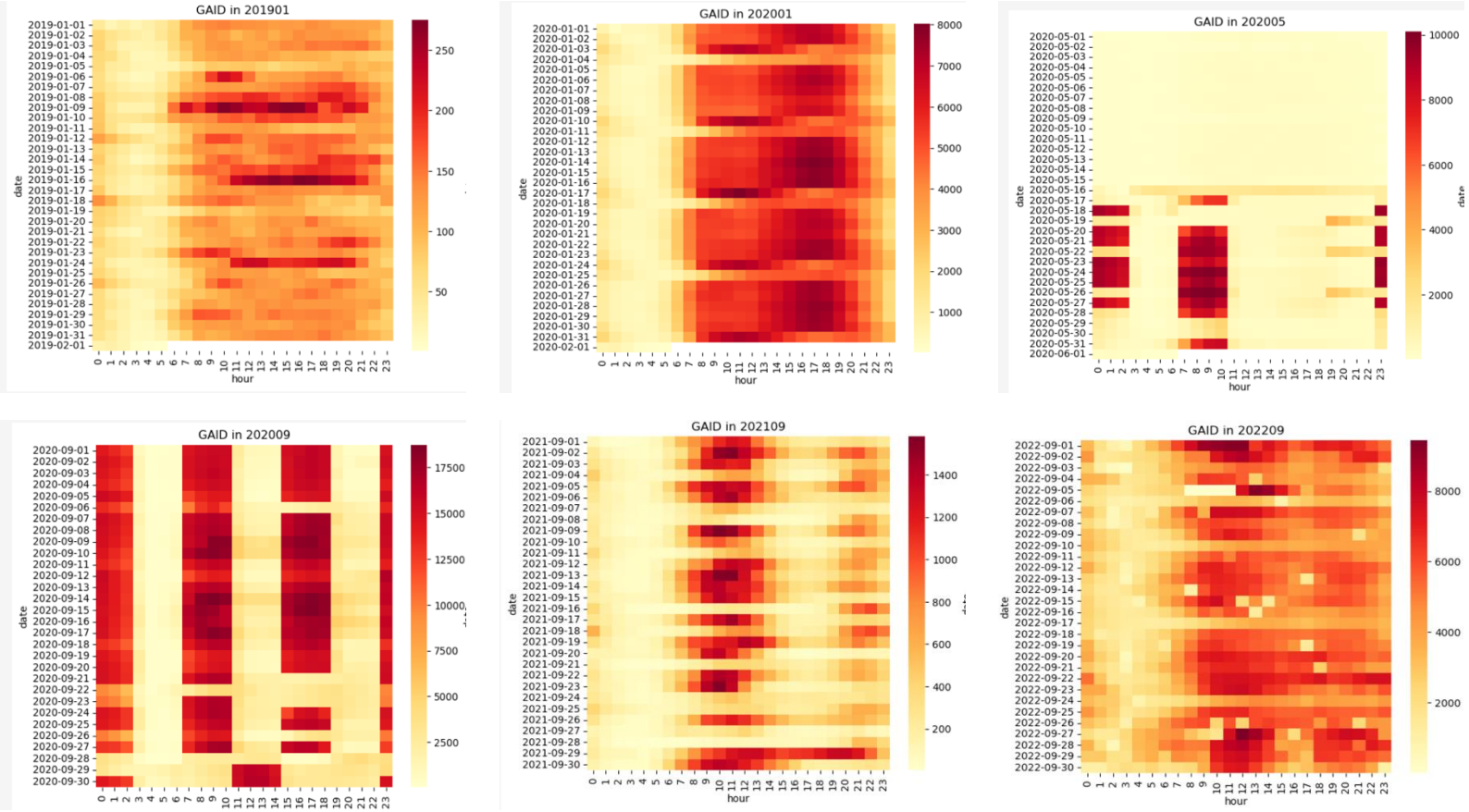
- 400 M total dwells/ stays
- 01/2020 - 09/2023
- ~200 000 users monthly (~4% of TAM)
- ~4 dwells per user per day
- Signals aggregated to visits of length >=10 min

Identifier	Identifier type	Local date time	Duration seconds	Centroid latitude	Centroid longitude	Bump count
001b3***115	GAID	01/01/2020 20:27:10	12036	31.7969337	34.70179411	14
0050c***f20	IDFA	01/01/2020 20:19:06	11218	31.8041044	34.76149723	19

Source: Habidatum

Signals on 10/01/2020

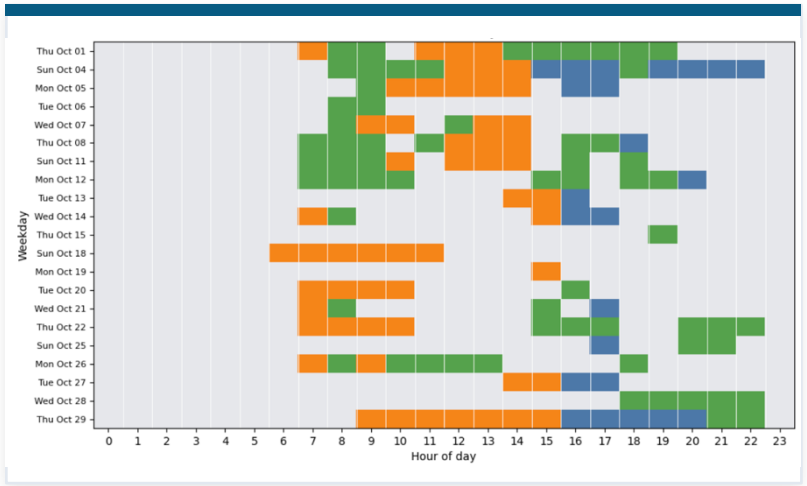
Signals are not evenly distributed in time



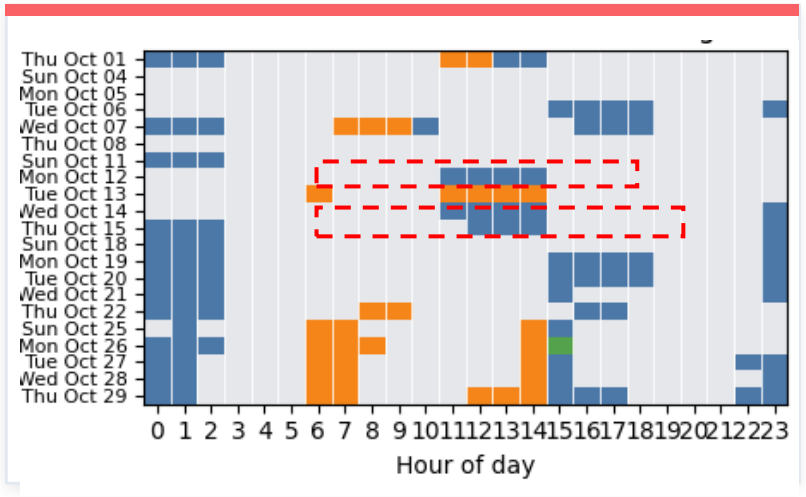
Frequency of signals changes between months, days and hours

User types based on work schedule

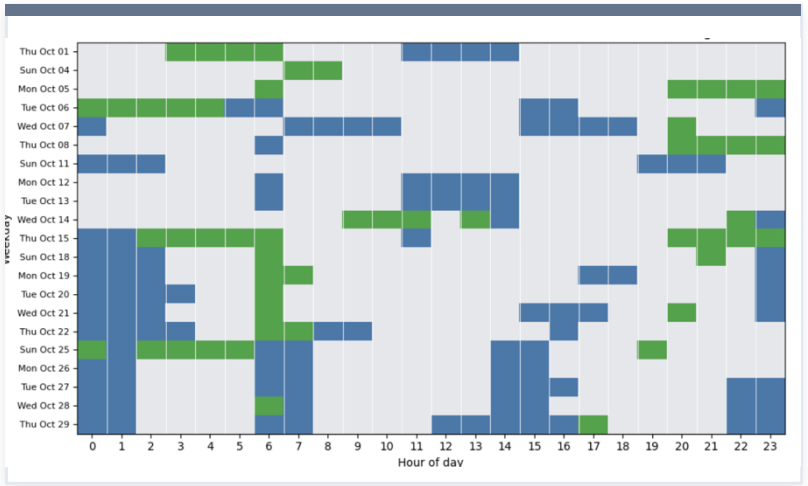
Full-time worker



Remote worker



Unknown



Weekday hourly locations for three work profiles, October 2020

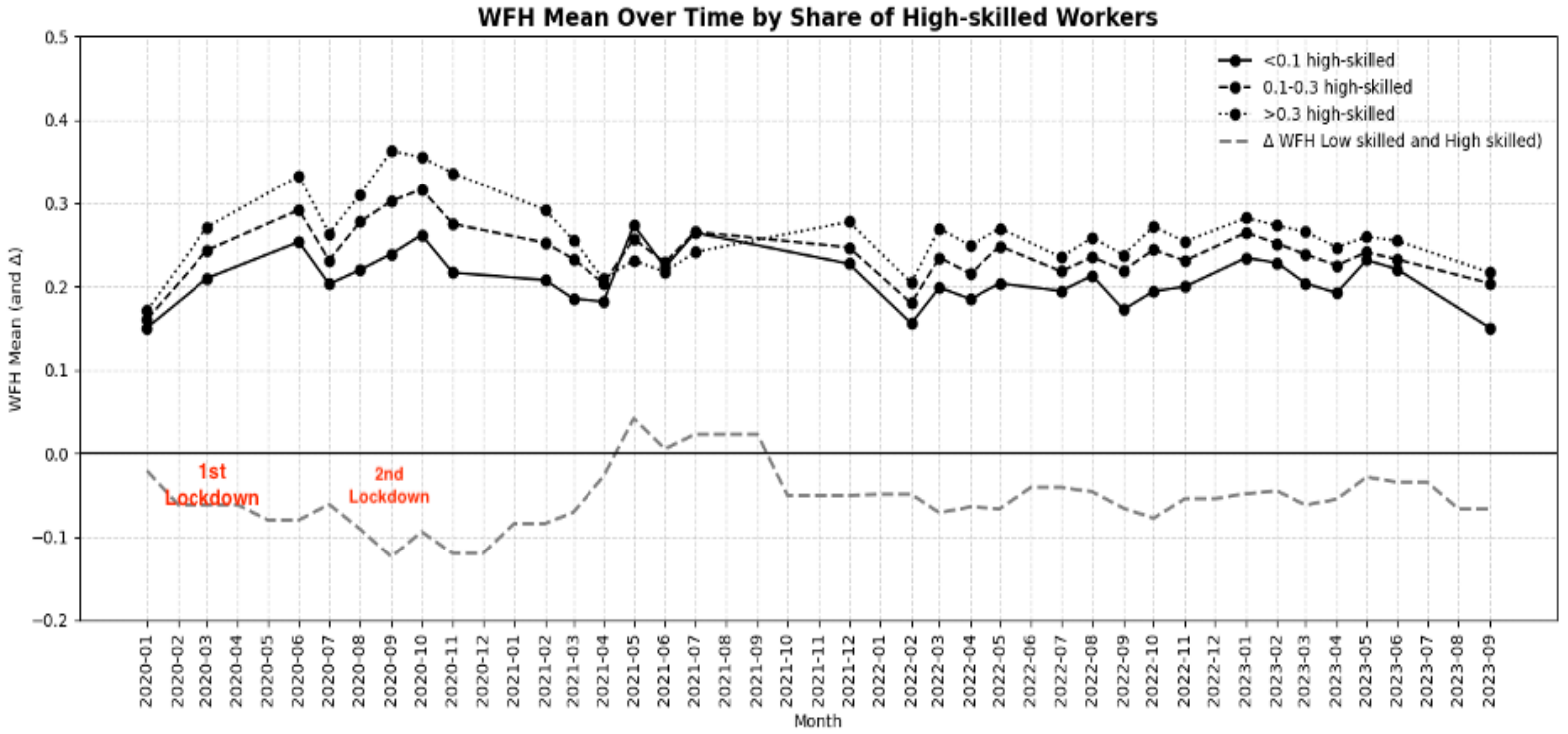


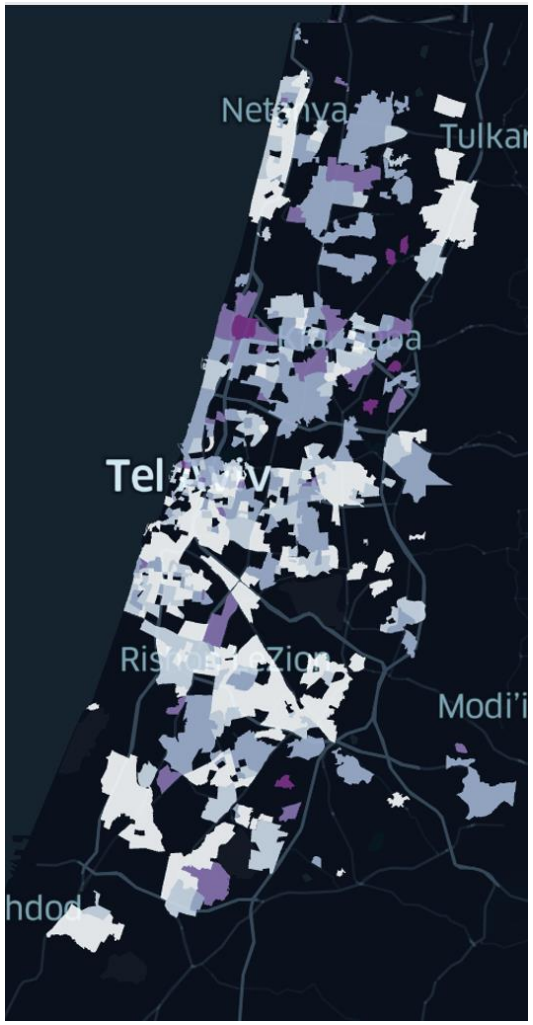
FIND OFFICE HOURS SPENT AT HOMEUSING BAYESEAN PROBABILITY

1 $P_i(W|h) = P_i(W) \cdot P_i(h|W) / P_i(h)$

2 $WH = 1$ *if* $P_i(W|h) > P_i(H|h)$ *and* $P_i(W|h) > P_i(A|h)$, *otherwise* 0

Remote work estimations



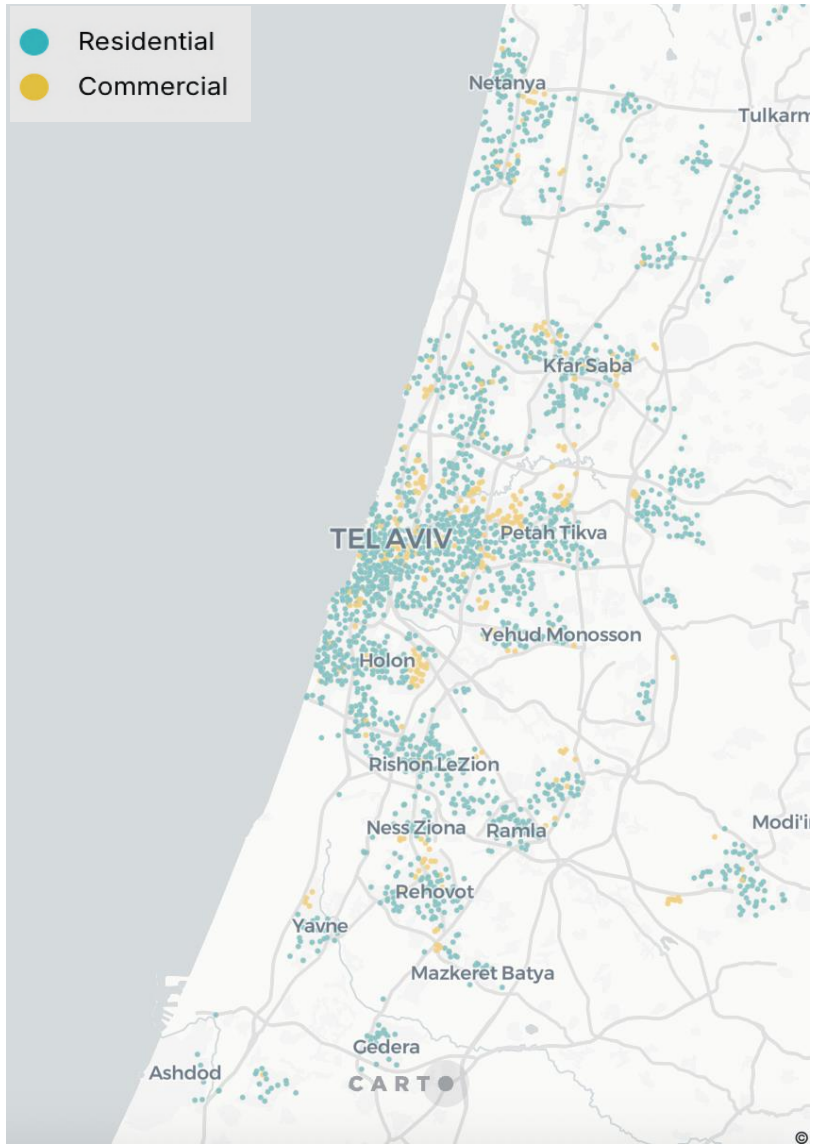


Remote work days share at 08/2023

Visits to third places

Points of interest · Google Places + OSM + Israel Census

CANDIDATE THIRD PLACES



CANDIDATE THIRD PLACES — **4,902**

Cafés, libraries, coffee shops, bakeries, museums, community & cultural centers, bookstores

Google Places API

WORK-ZONE & TRANSPORT HUBS — **-1,378**

Business centers, corporate offices, industrial areas, universities, schools

Google Places + OSM + Census

FINAL CANDIDATE THIRD PLACES — **3,524**

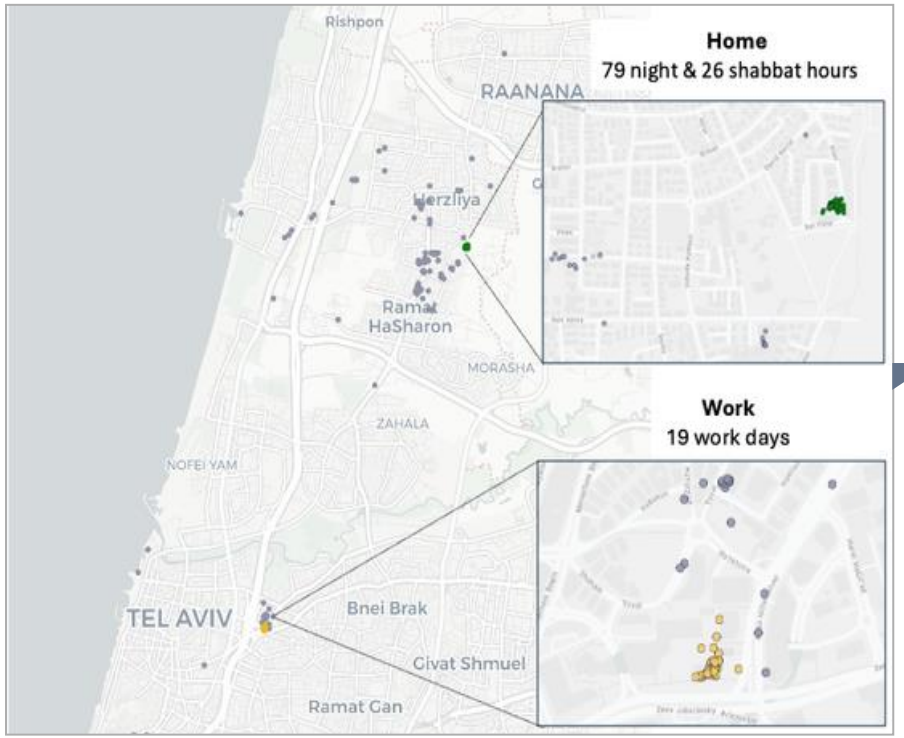
OTHER AMENITIES — **~34K**

Hospitals, Parks, Retail, Entertainment, Outdoor attractions, Lifestyle services

OpenStreetMap

Steps to classify visits on home/work/POI and other

1. Detecting users **home/work** monthly



Home:
>= 2 hours in 8pm -9 am
>=1 signal on Shabbat

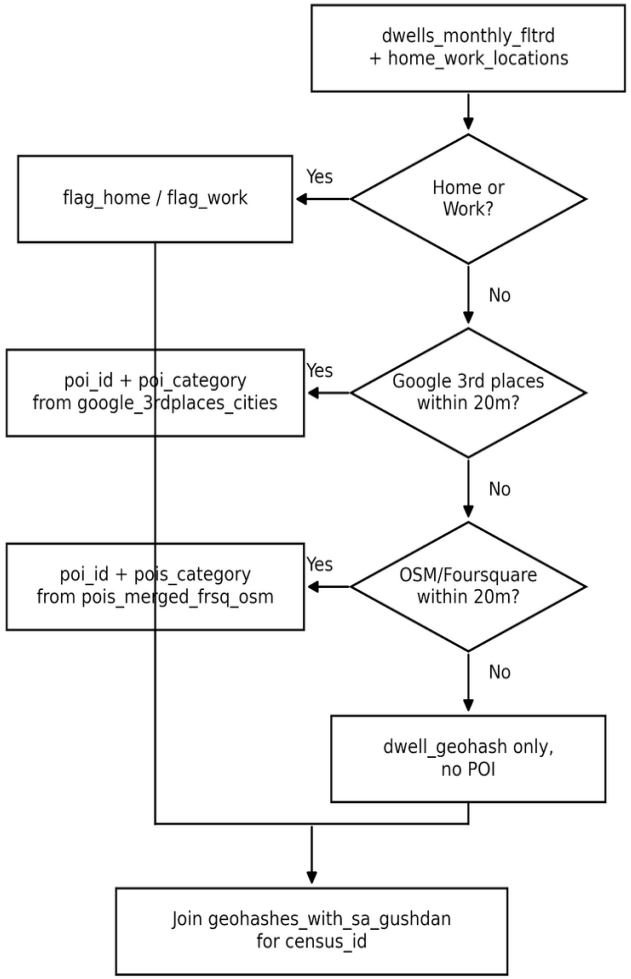
Work:
0 weekends
>=3 hours & 2 days
>=1km from home

2. Detecting **POIs** visits



Closest POI within 20m

Finalizing type of visit



METHODOLOGY · POSITIVE LABELS

Identifying third places using Google reviews

Review-derived features

-Number of reviews -Average Rating -Price range -Serve coffee -Restroom

Direct evidence

✓ INCLUDE

'laptop'
'place to/for work'
'remote work'
'business meeting'
'computer'
' get the work done '

✕ EXCLUDE

'event venue'
'no Wi-Fi'
'no laptop'
'on the go'
'repair'

Flag 'Suitable ambience'

✓ INCLUDE

'quiet'
'relaxing'
'Comfortable'
'calm'
'meeting friends'
'sit with'
'wifi'
'charger'

✕ EXCLUDE

'not quiet'
'rush'
'loud'
'not comfortable'
'not relaxing'
'not calm'

100 *places with explicit 'work-from-here' evidence*

Classification model

C

PU learning (Elkan – Noto)

Elastic-net logistic regression with PU adjustment.

Trusted positives only — 'unlabeled' rest

7 features × 3,524 candidates

100 iterations × 6 hyper-parameter grids

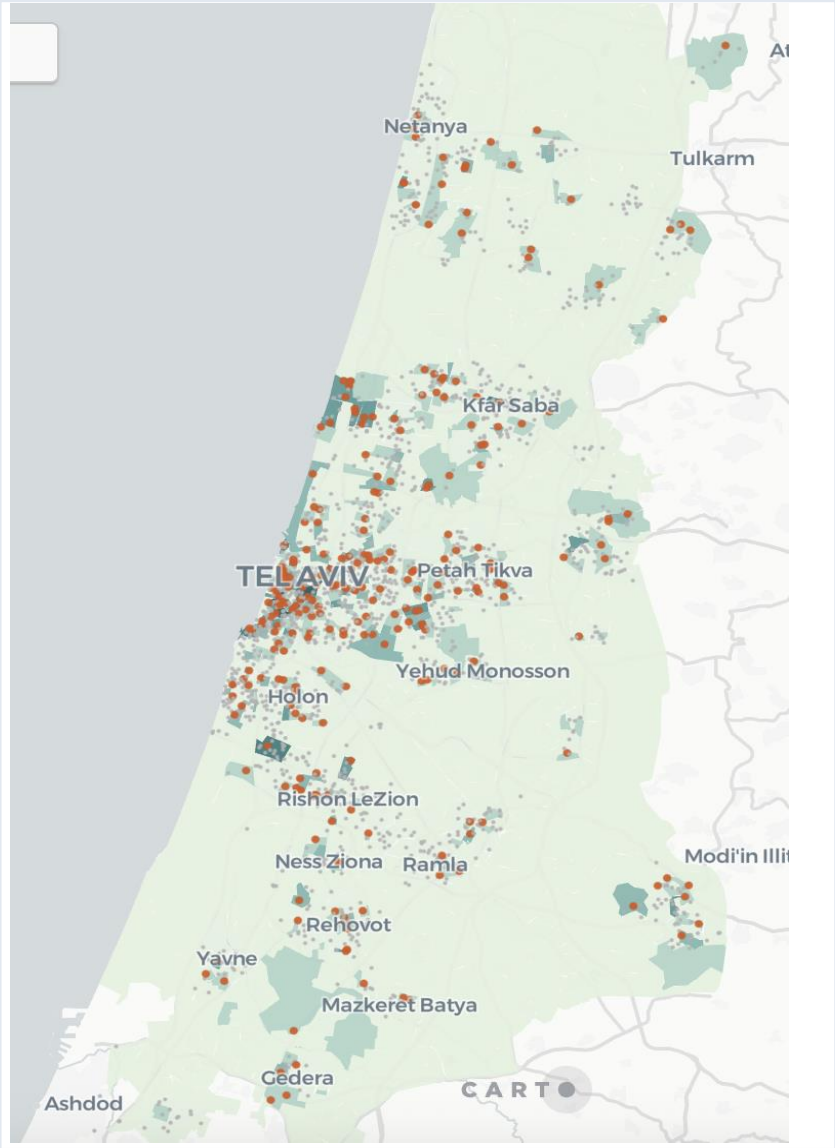
Validate: hide 30 % positives, recall@k

$$P(y=1|x) = P(s=1|x) / c$$

Ranked probability output

Detected places suitable for remote work

Identification Results · 550 new third places identified



550
places identified

15.6%
of all candidates

Tel Aviv CBD
coffee_shop **66 (34%)**

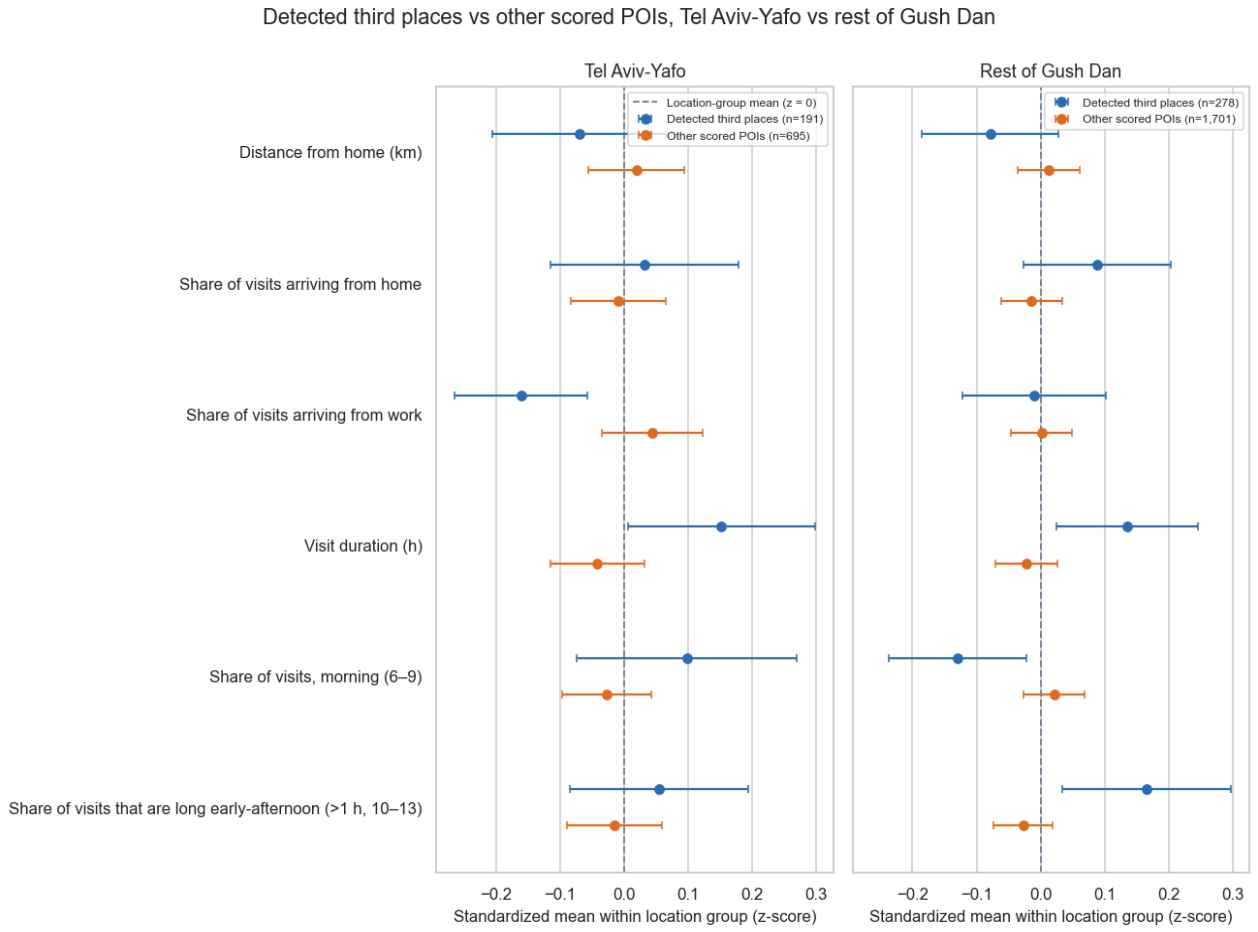
Tel Aviv Residential
Cultural centers **6 (31%)**

DETECTED BY POI TYPE (TAM)

POI TYPE	FOUND	% TOTAL	RECALL
coffee_shop	**174**	29.9 %	71 %
cafe	**128**	20.6 %	50 %
library	**50**	34.5 %	100 %
Cultural center	**62**	15.8 %	100 %
bakery	**66**	5.9 %	33 %
Community center	**27**	11.1 %	—

Visitors Behavior. How do visits differ between third places and the rest?

STANDARDIZED BEHAVIOR DIFFERENCES



SIGNIFICANT DEVIATIONS

Longer visits + 0.15 σ

All Tel Aviv Metropolitan area

Distance from work – 0.16 σ

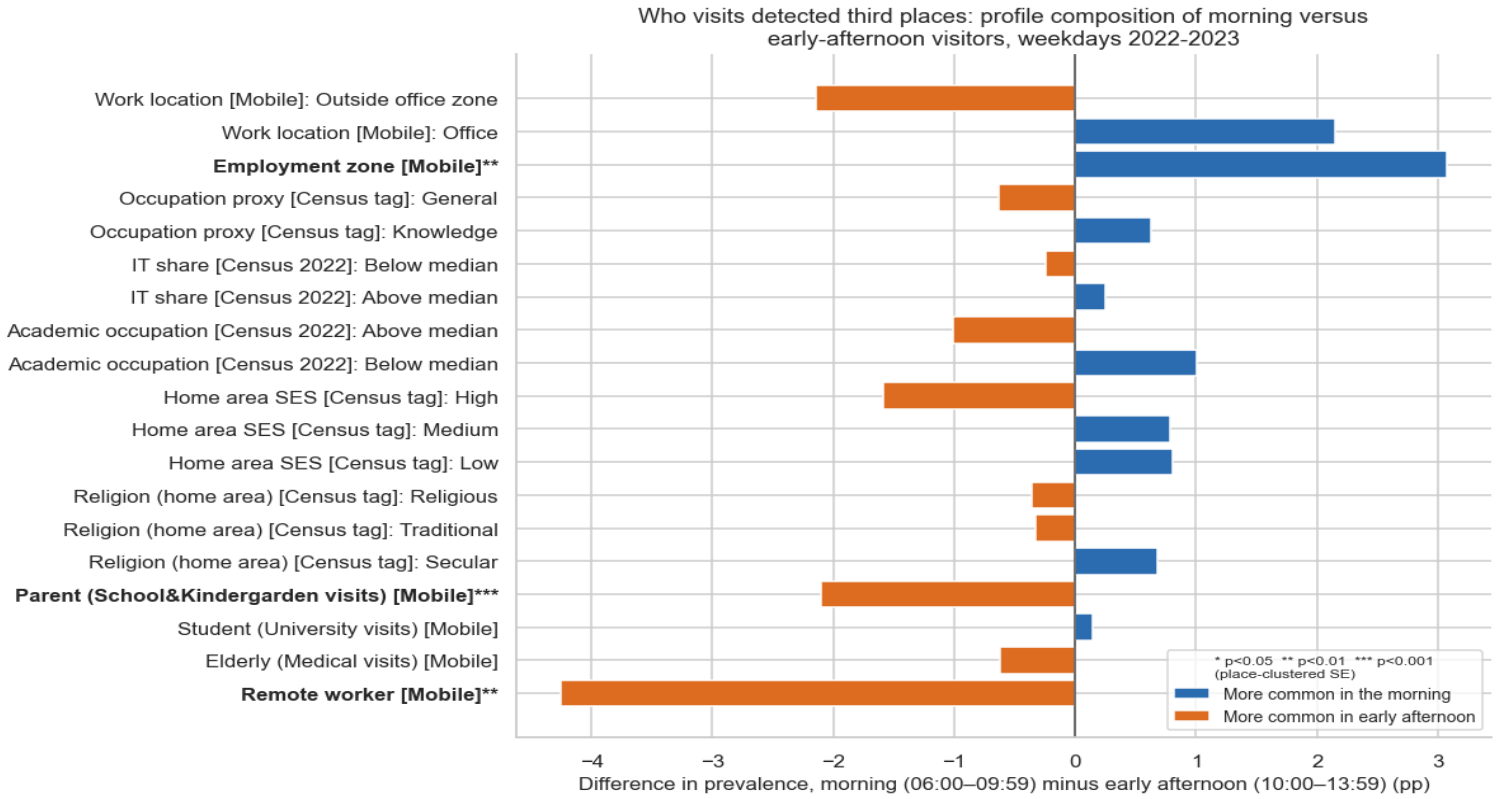
Only Tel Aviv

More long afternoon stays + 0.17 σ

Only satellite cities

Who are the visitors of the third places?

Prevalence of profile groups among morning visits (06:00 – 09:59) minus early-afternoon visits (10:00 – 13:59).



SIGNIFICANT GAPS

Remote workers ** — − 4 pp

Strongly under-represented in the morning window — they arrive in the early afternoon.

Parents on school run *** — − 2 pp

Strongly under-represented in the morning window — they arrive in the early afternoon.

Employment zone workers ** — + 3 pp

Office-anchored users pass through third places on their way to work in the morning.

OLS of group indicator on a morning dummy · SE clustered at place (poi_id) · * p<0.05 ** p<0.01 *** p<0.001

What GPS data can and cannot tell us

1 Granularity

GPS signals allow a temporal and spatial granularity of insights that is not available with traditional sources.

2 No human bias

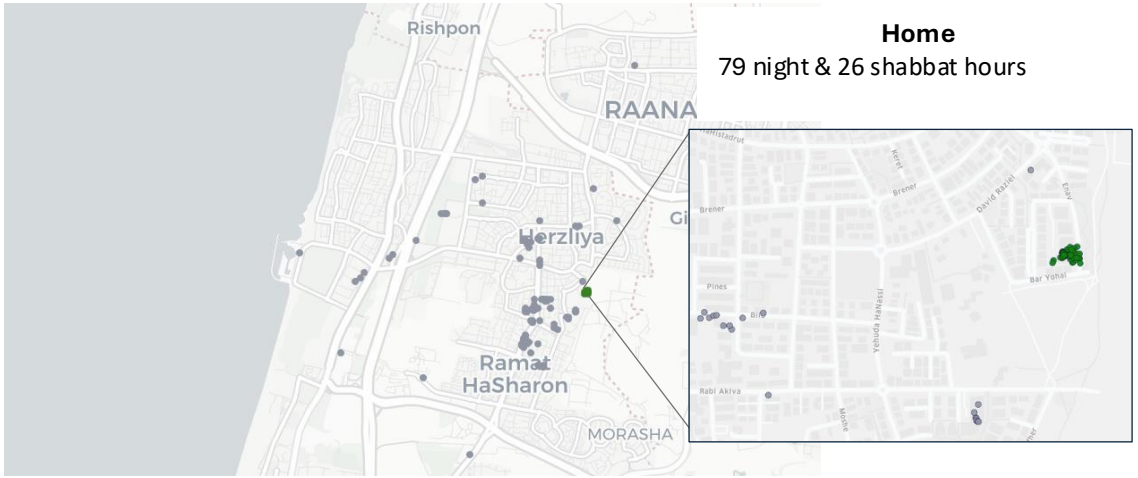
Changes in residential dynamics can be understood without asking residents directly.

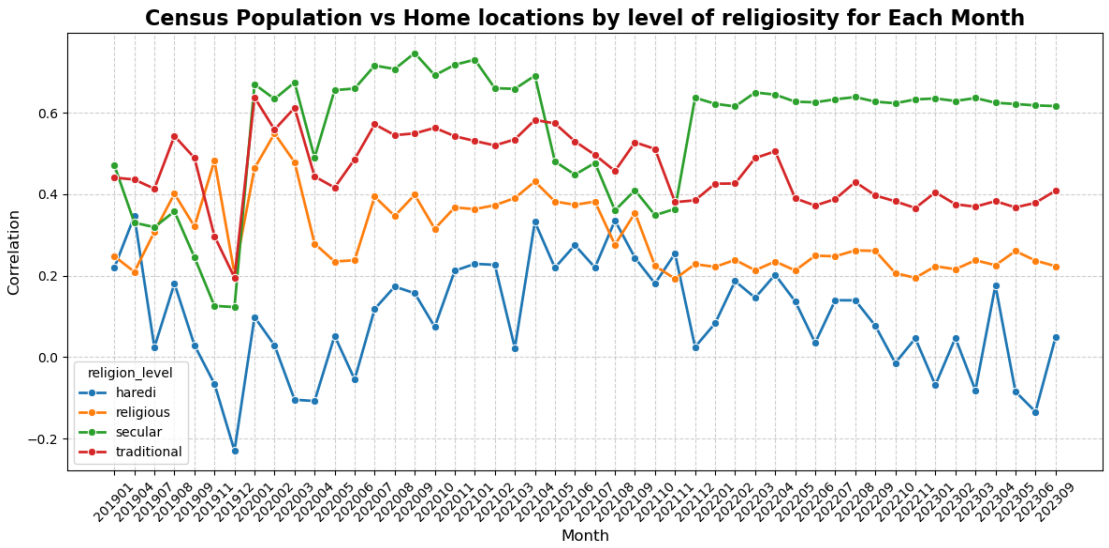
3 Not equally representative

The data naturally inherits biases that have to be considered: groups with lower smartphone use, such as the Haredi population, are underrepresented.

Appendix

Identification of home locations





CRITERIA

Home location detection

- At least 2 hours between 8 pm and 9 am
- At least 1 signal on Shabbat
- The place with the highest number of night + Shabbat signals that satisfies these criteria

VALIDATION

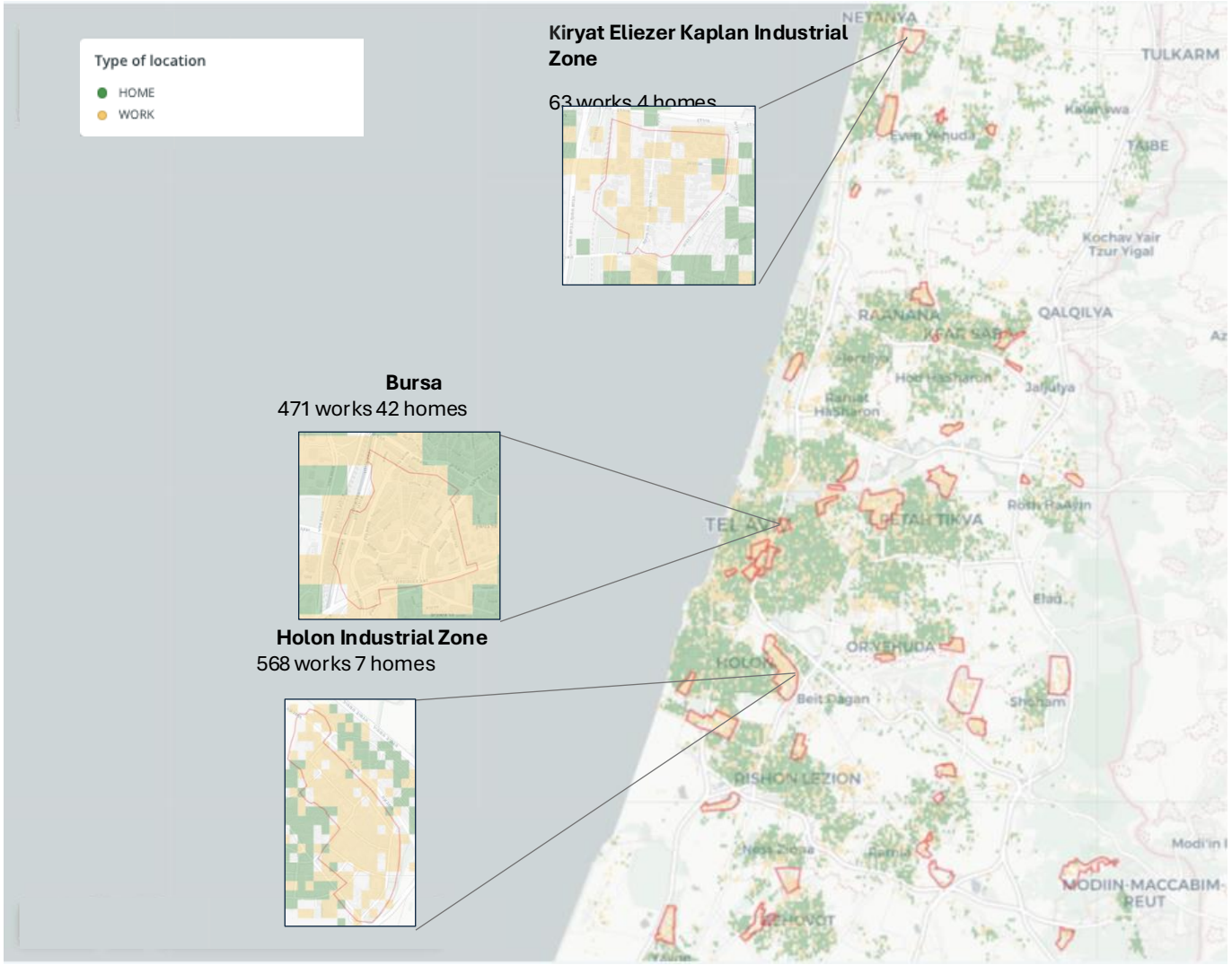
0.57

monthly correlation with census population across statistical areas

< 2 %

of homes attributed to a non-residential building

Detected work locations vs real work zones



CRITERIA

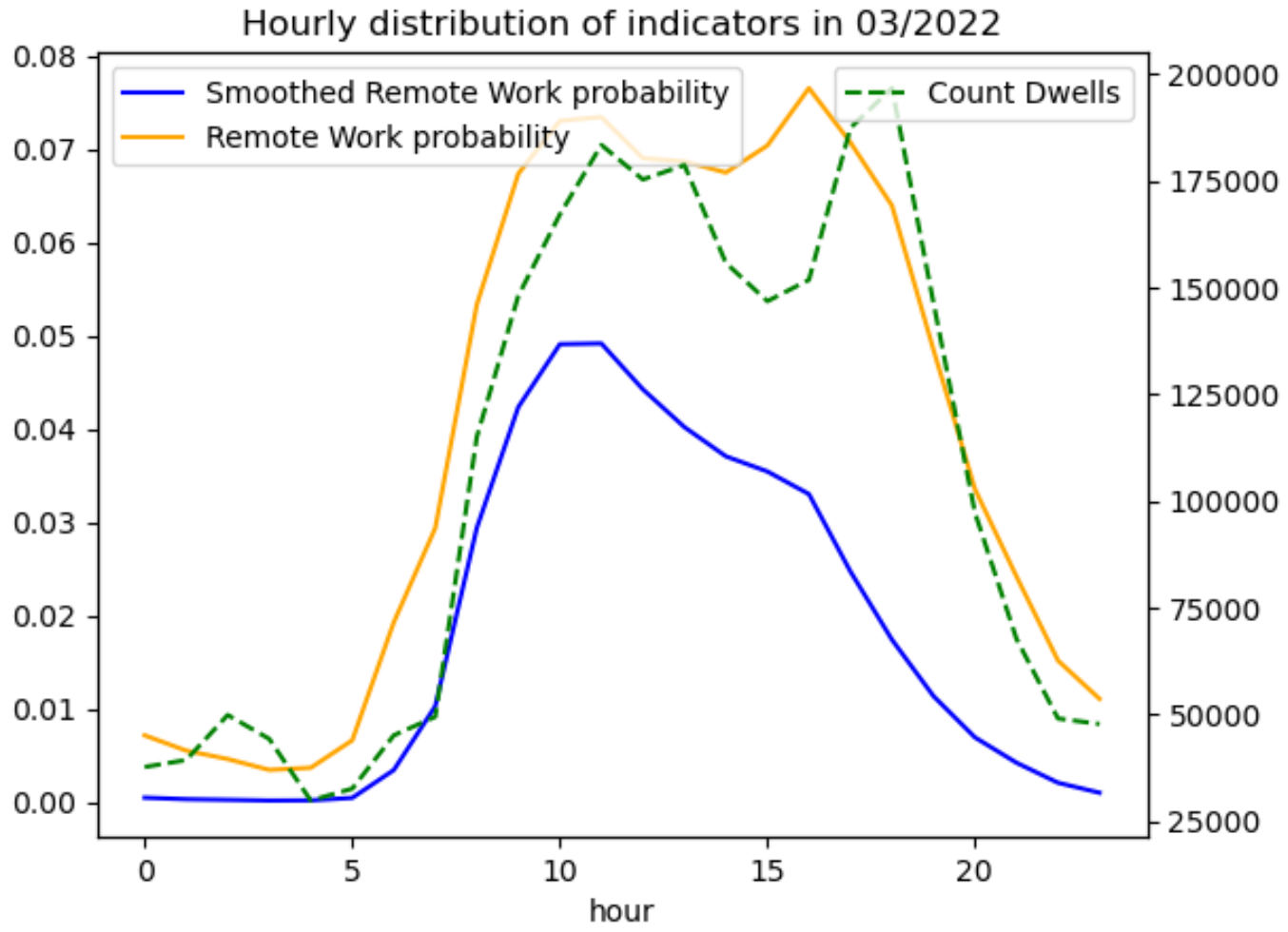
Work location detection

- No signals on weekends
- At least 3 hours and two days
- At least 1 km from home
- The place with the highest number of signals that satisfies these criteria

WORK LOCATION VS REAL WORK ZONES

Real work zone	Estimated work location	Share
FALSE	FALSE	69%
	TRUE	31%
TRUE	TRUE	96%
	FALSE	4%

Improvement: smoothing conditional probability



Smoothed remote work = Flag Remote work × P (W)